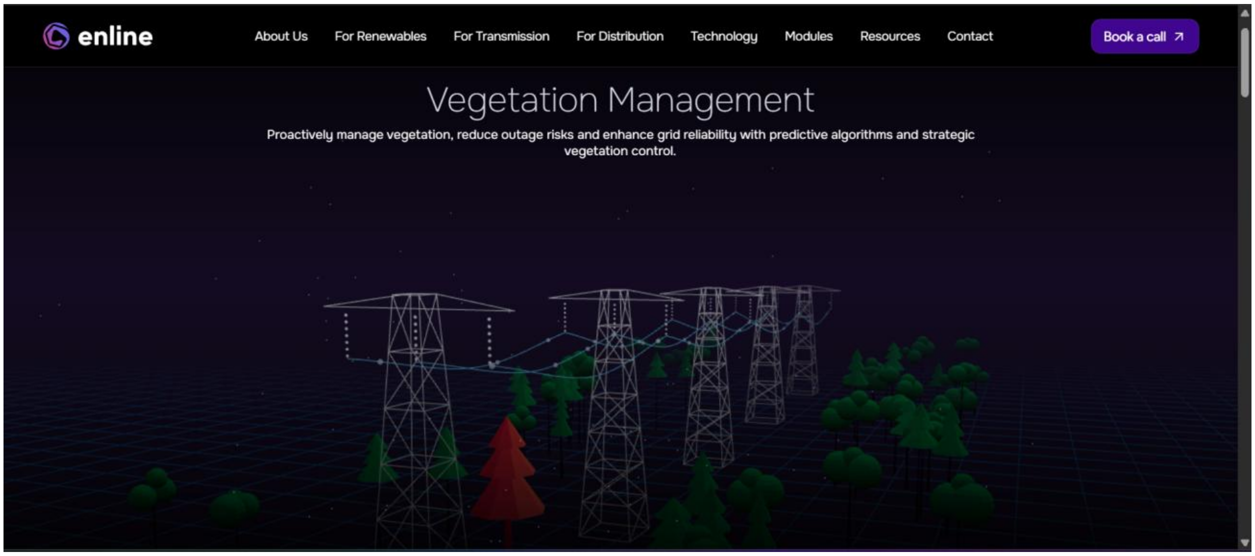
Enline Energy is an **AI-powered Digital Twin platform**

Enline Energy is an AI-powered Digital Twin platform that creates a real-time virtual replica of the power grid by combining physics-based models, AI, weather data, SCADA, and **existing grid information**. It provides utilities with a complete view of the network to monitor grid conditions, optimize capacity, integrate renewable energy, and reduce outages and operational costs. Unlike traditional systems, Enline uses a software-first, sensorless approach, relying on AI and physics models instead of installing large numbers of physical sensors, making it faster to deploy and more cost-effective while supporting smarter grid operations and decision-making.

Main Features

- **Dynamic Line Rating (DLR):** Safely unlocks up to 20-38% more transmission capacity from existing lines . It continuously calculates a line's safe capacity using real-time and forecasted weather conditions instead of relying on conservative static ratings . In a project with Red Eléctrica de España, Enline's DLR achieved over 95% accuracy compared to physical sensors and delivered an average power increase of 21.1% .
- **Network State Estimation:** Provides complete grid visibility by filling data gaps where telemetry is sparse . Uses physics-based models and AI to estimate the grid's state, enabling safer operations and faster fault detection . This capability is central to Enline's ability to deliver 100% transmission line visibility .
- **Power Optimization (OptiMax):** Optimizes the dispatch of active and reactive power in real-time to minimize electrical losses and maximize the efficient use of renewable energy . This feature helps reduce the curtailment of wind and solar power .
- **Topology Control:** Recommends optimal switching configurations for the grid using a deep reinforcement learning agent . This helps maximize grid flexibility, security, and resilience . When combined with other features, it contributes to a 22-35% reduction in congestion costs .
- **Vegetation Management:** Combines the digital twin with predictive analytics to identify high-risk vegetation growth hotspots near power lines . This allows utilities to move from fixed-cycle trimming to targeted, risk-based maintenance, improving safety and reducing costs .



https://enline.energy/articles/digital-twin-technology-for-power-grids

https://enline.energy/solutions/renewables

https://youtu.be/Nj6f2bLtEAo

The GE Vernova Grid Solutions website

The GE Vernova Grid Solutions website presents GridOS, an AI-powered software platform for modern power grid management. It introduces the company's vision for building a resilient and intelligent grid, explains the core GridOS platform, and provides detailed pages for solutions such as DERMS, ADMS, and AEMS. The website also includes technical blogs, product updates, customer success stories, industry achievements, and calls-to-action for learning more or contacting experts.

GridOS Features

- **Visual Intelligence (3D Digital Twin):** Creates a living, 3D digital twin of the entire grid from high-precision LiDAR data, providing operators with a real-world, continuously updated view of asset location and physical condition.
- **AI-Powered Grid Analytics & Optimization:** Uses AI and machine learning to automate complex tasks such as DER scheduling and voltage control, transforming raw data into predictive and automated actions for grid stability.
- **DER & Renewable Energy Management:**

    - **DERMS (Distributed Energy Resource Management System):** Orchestrates the full lifecycle of distributed energy resources, from integration to control and optimization.
    - **ADMS (Advanced Distribution Management System):** Ensures distribution grid reliability by balancing loads and integrating DERs in real-time.
    - **AEMS (Advanced Energy Management System):** AEMS manages the transmission grid by balancing power, maintaining grid stability, and integrating large-scale renewable energy sources.
- **Predictive Maintenance & Outage Prediction:** Enables proactive identification of grid disturbances and faster response, leading to quantifiable results like 21% fewer outages and 17% faster restoration.
- **Grid Resilience & Real-Time Monitoring:** Provides a unified view for real-time monitoring, optimization, and control, helping utilities prepare for, manage, and recover from disruptive events like severe storms.
- **Power Flow Analysis:** Performs critical network analysis based on a unified, governed data model, ensuring planners and operators work from the same trusted view of grid topology and power flow.

https://youtu.be/LO4w40ya0Ms

https://www.gevernova.com/software/products/gridos-for-distribution

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KOPR Smart Grid AI Twin (DataThings)

KOPR Smart Grid AI Twin is an AI-powered Digital Twin platform for electrical distribution networks. It creates a real-time virtual replica of the power grid by integrating data from GIS, ERP, SCADA, smart meters, IoT sensors, and other operational systems into a single unified model. The platform combines Artificial Intelligence, real-time monitoring, power flow analysis, and scenario simulation to help utilities monitor the grid, predict future conditions, optimize operations, and prevent outages before they occur. Its goal is to improve grid reliability, operational efficiency, and decision-making by providing operators with one intelligent platform instead of multiple disconnected systems.

Main Features

- **Digital Twin Model:** Builds a continuously updated virtual copy of the entire electrical grid by integrating data from multiple systems.
- **Data Consistency & Validation:** Detects and resolves inconsistencies between different data sources to ensure accurate and reliable grid information.
- **AI Consumption & Production:** Uses machine learning to forecast electricity demand, renewable generation, and identify potential overload hotspots.
- **Power Flow Analysis:** Calculates real-time voltage, current, power flow, transformer loading, and network losses across the grid.
- **What-if Scenario Simulation:** Safely evaluates future scenarios such as new EV chargers, solar installations, maintenance operations, or network reconfigurations before applying them to the real grid.
- **Prescriptive Decision Support:** Recommends operational actions—such as switching feeders, redistributing loads, or controlling smart devices—to prevent overloads and outages.
- **Customizable Platform:** Allows utilities to deploy only the features that match their operational needs and digital maturity.

For example:

- A **small utility** may only want:
  - Digital Twin
  - AI demand forecasting
  - Dashboard
- A **large utility** may need:
  - AI forecasting
  - Digital Twin
  - Power flow analysis
  - Scenario simulation
  - Decision support
  - Data validation

https://kopr-twin.com/

https://youtu.be/or3XSWjht3Y

1) **Schneider Electric EcoStruxure Grid Operation**

   Website: Schneider Electric EcoStruxure Grid Operation

   Video1: https://www.youtube.com/watch?v=CgffDiN3tF4

   Video 2: https://www.youtube.com/watch?v=2cOJ240GYuw

   Demo-video: https://youtu.be/R2HuFP-7sAI?si=CMrZsasI1VdLxwTA

Why did Schneider build EcoStruxure Grid?

Imagine you are responsible for supplying electricity to an entire city.

Every second you need to answer questions like:

- Is every neighborhood receiving electricity?
- Did a transformer fail?
- Is a transmission line overloaded?
- Is the solar farm producing enough electricity?
- Should batteries start supplying power?
- Which customers lost electricity?

Years ago, operators had many separate software systems:

- One for monitoring.
- One for outages.
- One for maps.
- One for alarms.
- One for field engineers.

It was slow and confusing.

So Schneider created EcoStruxure Grid Operation, which combines everything into one platform.

What does the software actually do?

Think of it as the control room for the electrical grid.

Imagine this dashboard:

GRID STATUS

🟢 Solar Farm

Working

🟢 Wind Farm

Working

🟢 Power Plant

Working

🟡 Battery

Charging

🔴 Transformer #12

Overheating

Electricity Demand

785 MW

Renewable Energy

46%

Active Alerts

3

Instead of checking hundreds of devices manually, operators can see everything on one screen.

Feature 1 — Real-Time Monitoring

The software receives live information from thousands of sensors installed across the electrical grid.

It continuously shows:

- Power plants
- Solar farms
- Wind farms
- Batteries
- Transformers
- Substations
- Power lines
- Customer demand

The information updates every few seconds.

Feature 2 — SCADA

You will often hear the term SCADA.

It stands for:

Supervisory Control and Data Acquisition

Think of SCADA as the "eyes and hands" of the electrical grid.

It:

- Collects information from sensors.

- Displays everything on the dashboard.
- Allows operators to remotely control equipment.

For example:

Instead of sending an engineer to turn off a switch,

the operator clicks a button,

and the switch changes remotely.

SCADA makes the grid much easier to manage.

Feature 3 — Digital Twin

Schneider also has an operational Digital Twin.

Imagine a virtual copy of the electrical grid.

Operators can:

- View the current network.
- Understand how electricity is flowing.
- Test changes before applying them.

For example:

"What happens if this transmission line fails?"

The Digital Twin can simulate the answer first.

Feature 4 — Outage Management

This is one of Schneider's strongest features.

Imagine lightning strikes a power line.

Without software:

People call the electricity company.

Engineers try to discover where the problem is.

This may take hours.

With EcoStruxure:

The software automatically:

- Detects the outage.
- Finds the likely fault location.
- Estimates which customers are affected.
- Helps dispatch repair crews.
- Suggests the fastest restoration plan.

Feature 5 — Smart Alarms

Instead of simply saying:

"Something is wrong."

The software can generate useful alerts like:

Transformer temperature too high.

Line overloaded.

Communication lost.

Equipment not responding.

Operators know exactly where to investigate.

Feature 6 — Mobile Application

Field engineers don't have to keep calling the control room.

They receive:

- Maps

- Work orders
- Equipment status
- Network changes

directly on their mobile devices.

This makes repairs faster and reduces mistakes.

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3) ABB Ability™ Network Manager

- • Real**-Time Grid Monitoring** – Monitors power plants, substations, transmission lines, renewable energy, and grid status in real time.
- **SCADA** – Collects live data from the grid and allows operators to remotely monitor and control electrical equipment.
- **Network Analysis** – Analyses power flow, voltage, and grid conditions to identify potential issues.
- **Outage Management** – Detects faults, locates outages, identifies affected customers, and helps restore power quickly.
- **Decision Support** – Recommends actions such as load transfer, network reconfiguration, and fault isolation to assist operators.
- **Operator Training Simulator** – Provides a virtual environment for operators to practice handling faults and emergency scenarios.
- **Interactive Dashboard** – Displays the electrical network, alarms, equipment status, and power flow on one screen.
- **Renewable Energy Integration** – Supports monitoring and management of renewable energy resources within the grid.
- **Cybersecurity & Reliability** – Includes secure communication and tools to improve grid reliability and operational resilience.